Meat, Fish & Animal Products Preservation Field Manual

12 methods for preserving meat, fish, and animal products at home

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Preserving plant food is mostly about stopping rot. Preserving animal food is about stopping rot and preventing something significantly more dangerous — botulism, trichinosis, and a short list of other hazards that can cause serious illness or death without any visible or odor warning.

That's not a reason to avoid preserving meat, fish, and animal products. It's a reason to understand the science before you start. Every method in this guide has been used by ordinary people for generations. Every number — cure ratio, processing time, internal temperature, water activity — is here because it matters to your safety.

This guide covers twelve methods for preserving meat, fish, poultry, rendered fats, and other animal products at home, without commercial processing equipment. Methods range from ancient (salt fish, pemmican) to modern (pressure canning). Some require significant investment in equipment; some require almost nothing. All of them work.

How to Use This Guide

Each method section covers what it works for, what it doesn't, complete equipment, step-by-step process with specific temperatures and ratios, critical safety information, storage duration, and spoilage recognition.

Start with what you have. If you have a freezer, start there. If you have a pressure canner, you can preserve almost anything. If you're building toward more traditional methods — curing, smoking, drying — read those sections fully before purchasing equipment.

On curing salts: Several methods in this guide require sodium nitrite (Prague Powder #1) or a nitrite/nitrate blend (Prague Powder #2). These are not optional for the products that call for them. They prevent botulism in the anaerobic interior of cured meat. Using plain salt where curing salts are specified is not a substitution — it's a safety failure.

Methods covered in the companion guide: Basic freezing techniques (including blanching vegetables before freezing), water bath canning, lacto-fermentation of vegetables, and vinegar pickling are covered in the *Harvest Preservation Field Manual*. This guide covers animal products specifically.

Important Notice

Preserving meat, fish, and animal products carries higher safety stakes than preserving vegetables. Botulism, trichinosis, and other pathogens can cause serious illness or death without any visible or odor warning. Where this guide specifies curing salts (Prague Powder #1 or #2), they are not optional — they prevent botulism in cured products. Follow every temperature, ratio, and processing time exactly as written. pH testing is mandatory for any canned product. When in doubt about any preserved animal product, discard it. The cost of a lost jar is nothing compared to the alternative.

Part One: Equipment

Level 1: Freezing and Basic Preservation

What you likely already have, or can acquire for under $50.

- Chest or upright freezer at 0°F or below — the single most important piece of preservation equipment for animal products. A chest freezer is more energy-efficient and holds temperature longer during power interruptions.
- Heavy-duty zip-lock freezer bags — gallon and quart sizes; label everything with contents and date
- Permanent marker
- Digital instant-read thermometer — mandatory; visual doneness is not safe for meat
- Kitchen scale — required for accurate cure ratios; volume measurements for salt are unreliable
- Large stockpot (8+ quart) for making stock and blanching

Level 2: Canning, Curing, and Dehydrating

One-time investments that open shelf-stable preservation.

- Pressure canner (16+ quart minimum) — see Method 1 for type selection. This is the single most critical equipment investment for shelf-stable animal product preservation.

- Presto 23-quart (dial gauge): least expensive; requires annual gauge calibration
- All American 21.5 or 25-quart (weighted gauge): more expensive; no calibration required; metal-to-metal seal (no gasket to replace)

- Mason-style canning jars (Ball, Kerr, Bernardin) — pint and half-pint sizes are most important for meat and fish canning. Quart jars are used for stock and poultry only.
- New two-piece lids each season — flat discs are single-use; bands are reusable
- Jar lifter, canning funnel, bubble remover
- Food dehydrator — 4–9 tray, thermostat to 160°F, with fan. Required for jerky made safely without

pre-cooking.

- Non-iodized pickling/canning salt — the foundation of every brine and cure
- Prague Powder #1 (Cure #1 / InstaCure #1) — 6.25% sodium nitrite; for short-cure and cooked/smoked products
- Prague Powder #2 (Cure #2 / InstaCure #2) — for long dry-cured and fermented products
- Meat injector — for bone-in hams and thick roasts where surface cure alone won't penetrate adequately
- Probe thermometer (leave-in type) — for monitoring internal temperature during smoking and curing
- Food-grade storage containers — ceramic crocks, glass carboys, or food-grade plastic buckets for brining and barrel curing

Level 3: Smoking and Sausage

- Smoker — electric (most controllable for beginners), pellet, or offset barrel. An electric smoker with a precise temperature controller is the safest option for meat that must reach specific internal temperatures.
- Sausage stuffer — 5-quart horizontal or vertical stuffer for hand-stuffed sausages
- Meat grinder — #12 or larger; cast iron hand-crank or electric
- Natural casings — hog (32–35mm), sheep (22–24mm), or beef middles (40–60mm) depending on sausage type; packed in salt, refrigerated
- Vacuum sealer with bags — significantly extends quality of cured and smoked products
- pH meter (calibrated) — required if producing fermented dry sausage
- Water activity meter — required for fermented sausage to verify shelf stability; optional for other applications
- Curing chamber — a temperature and humidity-controlled environment for drying fermented sausage; a spare refrigerator with a humidity controller (Inkbird or equivalent) and a small USB fan is a common and functional setup

Part Two: Methods

Method 1: Pressure Canning — Meat and Poultry

Pressure canning is the only safe method for home canning any animal product. Boiling water reaches 212°F — insufficient to destroy Clostridium botulinum spores in a low-acid, protein-rich environment. A properly operating pressure canner at 10–11 PSI reaches 240–250°F, which destroys those spores.

What it works for:
Chicken, rabbit, duck, turkey (boneless or bone-in pieces), beef, pork, lamb, venison, bear, ground meat, stew meat, broth and stock.

What it does not work for:
Dairy products, eggs, butter, cream, or any recipe with added starch (flour, cornstarch) or pasta — these create density that prevents adequate heat penetration.

Equipment Setup

Add water to the canner per manufacturer instructions (typically 2–3 quarts of hot water). Load a jar rack. Prepare food and fill jars while the canner is heating.

The 10-minute vent — mandatory, every time:
After loading jars and locking the lid, leave the vent open. Heat on high until a steady, uninterrupted stream of steam has been exhausting for exactly 10 minutes. This purges trapped air from the canner. Air in the canner results in a lower actual temperature than the gauge reads — your food will be under-processed even if the gauge shows the correct pressure. Do not shorten this step.

After 10 minutes of steady venting: place the weight or close the petcock. Wait for the canner to reach required pressure. **Start timing only once the canner has reached correct pressure.** Maintain steady pressure throughout processing. Significant drops and spikes cause liquid to be pulled from jars.

When processing time is complete: turn off heat. Let pressure drop to zero naturally — do not force it. When gauge reads 0 PSI (or weighted gauge stops rocking), wait 10 additional minutes. Open vent; wait 10 more minutes. Open lid away from you. Remove jars straight up using a jar lifter.

Headspace and Salt

- Headspace: 1¼ inches for poultry; 1 inch for red meat and stock
- Salt: Optional for all meat canning — ½ tsp per pint, 1 tsp per quart. Salt is for flavor only, not safety.
- Remove excess fat from all meats before packing — fat on jar rims prevents proper sealing; fat in the jar can go rancid

Chicken and Rabbit

Preparation: Dress and chill chicken 6–12 hours before canning. Soak rabbit pieces 1 hour in salted water (1 tablespoon salt per quart), then rinse. Cut into pieces. Remove excess fat.

Two packing methods:

Hot pack (preferred): Boil, steam, or bake meat until about two-thirds done. Pack pieces into hot jars. Add hot broth, leaving 1¼ inch headspace.

Raw pack: Fill jars loosely with raw pieces. Do not add liquid. Leave 1¼ inch headspace. The meat creates its own liquid during processing.

Hot pack is preferred for better liquid coverage and storage quality.

Processing times:

Pack	Jar	Dial-gauge 0–2,000 ft	Weighted 0–1,000 ft
Boneless	Pints	75 min / 11 PSI	75 min / 10 PSI
Boneless	Quarts	90 min / 11 PSI	90 min / 10 PSI
Bone-in	Pints	65 min / 11 PSI	65 min / 10 PSI
Bone-in	Quarts	75 min / 11 PSI	75 min / 10 PSI

Beef, Pork, Lamb, Venison, Bear (Strips, Cubes, or Chunks)

Hot pack (preferred): Precook meat until rare by roasting, stewing, or browning in a small amount of fat. Pack pieces into jars; add boiling broth, meat drippings, water, or tomato juice. Leave 1-inch headspace.

Raw pack: Fill jars with raw pieces, 1-inch headspace, no added liquid.

Processing times:

Jar	Dial-gauge 0–2,000 ft	Weighted 0–1,000 ft
Pints	75 min / 11 PSI	75 min / 10 PSI
Quarts	90 min / 11 PSI	90 min / 10 PSI

Venison note: Adding one part high-quality pork fat to 3–4 parts ground venison before canning produces a significantly better texture in the jar.

Ground Meat

Shape into patties, balls, or 3–4 inch links. Cook until lightly browned. Remove excess fat. Pack into jars; add boiling broth, tomato juice, or water. Leave 1-inch headspace.

Processing times: Same as strips and cubes — 75 min/pints, 90 min/quarts at 11 PSI (dial-gauge, 0–2,000 ft).

Altitude Adjustments (Dial-Gauge)

Elevation	Pressure
0–2,000 ft	11 PSI
2,001–4,000 ft	12 PSI
4,001–6,000 ft	13 PSI
6,001–8,000 ft	14 PSI

Weighted-gauge: 10 PSI at 0–1,000 ft; 15 PSI above 1,000 ft.

Annual Dial Gauge Calibration

Bring your dial-gauge canner lid to your county cooperative extension office each spring. An inaccurate gauge means under-processing. Most extension offices test it for free. If the gauge reads more than 2 PSI off, replace it before canning. Weighted-gauge canners do not drift and do not require calibration.

Method 2: Pressure Canning — Fish and Seafood

Fish canning carries additional considerations beyond meat. *Clostridium botulinum* Type E — the strain associated with fish and marine environments — is cold-tolerant and can grow at temperatures as low as 38°F. Fish must be handled on ice from catch to jar. **Only pint jars are approved for home fish canning** — quart jars have not been tested and must not be used.

Fatty Fish: Salmon, Trout, Steelhead, Mackerel

Preparation: Eviscerate within 2 hours of catching. Keep on ice throughout. Remove head, tail, fins, and scales. Wash thoroughly and remove all blood. Cut into lengths that fit vertically in pint jars (approximately 3½ inches). Pack skin side next to glass.

Headspace: 1 inch. **Do not add liquid.** Add ½–1 tsp salt per pint (optional, flavor only).

Processing (raw pack, pint jars only):

Gauge type	Time	Pressure
Dial-gauge 0–2,000 ft	100 min	11 PSI
Dial-gauge 2,001–4,000 ft	100 min	12 PSI
Dial-gauge 4,001–6,000 ft	100 min	13 PSI
Dial-gauge 6,001–8,000 ft	100 min	14 PSI
Weighted-gauge 0–1,000 ft	100 min	10 PSI
Weighted-gauge above 1,000 ft	100 min	15 PSI

Note: White crystals may form in canned salmon (magnesium ammonium phosphate). These are harmless and usually dissolve when heated. They cannot be prevented.

Tuna (Unique Pre-Cook Requirement)

Tuna must be cooked before canning — the only home-canned fish that requires this.

1. Keep tuna on ice. Clean and remove viscera.
2. Pre-cook: Bake at 250°F for 2½–4 hours, or at 350°F for 1 hour, or steam 2–4 hours until internal temperature reaches 165–175°F.
3. Refrigerate overnight to firm the meat.
4. Remove skin, blood vessels, discolored flesh, all bones, and fin bases. Cut to jar length.
5. Fill jars tightly, pressing to a solid pack. Pack in water or oil. Leave 1-inch headspace.
6. Add 1 tsp salt per pint (optional).

Processing: 100 minutes at the same pressures as fatty fish above.

Clams

1. Scrub live clams; rinse; steam 5 minutes until shells open.
2. Extract meat; save juice.
3. Wash meat in salted water (1 tsp salt per quart of water).
4. Cover with boiling water containing 2 tablespoons lemon juice or ½ tsp citric acid per quart; boil 2 minutes; drain.
5. Fill jars loosely. Add hot clam juice plus boiling water to cover. 1-inch headspace.

Processing (hot pack):

Jar	Dial-gauge 0–2,000 ft	Weighted 0–1,000 ft
Half-pints	60 min / 11 PSI	60 min / 10 PSI
Pints	70 min / 11 PSI	70 min / 10 PSI

Oysters

Heat live oysters in a 400°F oven for 5–7 minutes until shells open. Cool briefly in ice water. Extract meat; wash in salted water (½ cup salt per gallon). Add ½ tsp salt per pint (optional). Fill jars with meat and hot water. 1-inch headspace.

Processing: 75 minutes at 11 PSI dial-gauge (0–2,000 ft), 10 PSI weighted. Half-pints or pints only.

Smoked Fish — Pressure Canning After Smoking

This produces a product with both smoke flavor and shelf stability.

1. Lightly smoke the fish for flavor — this is a short smoke, not a full hot smoke through.
2. Pack vertically in pint jars, 1-inch headspace, no added liquid.
3. Add exactly 4 quarts of cool tap water to the pressure canner (not less, not preheated).
4. Process 110 minutes at 11 PSI dial-gauge (0–2,000 ft), 10 PSI weighted (0–1,000 ft).
5. Pint jars only. Do not use a pressure saucepan — only 16–22 quart canners are validated for fish.

Method 3: Dry Salt Curing — Bacon, Ham, Salt Pork

Dry curing uses salt (and curing salts) applied directly to the surface of meat. Salt draws moisture outward via osmosis, lowering the water activity inside the muscle. Sodium nitrite, delivered through curing salts, penetrates the meat and inhibits Clostridium botulinum in the anaerobic interior.

Plain salt alone does not prevent botulism in the interior of a thick cut of cured meat. This is why curing salts exist.

Prague Powder: What Each One Does

Prague Powder #1 (Cure #1, InstaCure #1, Pink Curing Salt #1)

- Composition: 93.75% sodium chloride + 6.25% sodium nitrite
- Dyed pink to prevent confusion with plain salt
- Application rate: 1 oz (28g) per 25 lbs of meat. Approximately 1 level teaspoon per 5 lbs.
- For: Short-cure products that will be cooked or eaten within weeks: bacon, fresh ham, corned beef, pastrami, hot dogs, brined poultry.
- The nitrite in #1 breaks down during cooking and does not persist in the finished cooked product.

Prague Powder #2 (Cure #2, InstaCure #2, Pink Curing Salt #2)

- Composition: 89.75% sodium chloride + 6.25% sodium nitrite + 4% sodium nitrate
- Application rate: Same — approximately 1 oz per 25 lbs.
- For: Long dry-cured, air-dried, and fermented products that will NOT be cooked before eating: dry-cured whole hams, salami, lonza, coppa, dry-cured sausages with weeks-to-months cure times.
- The sodium nitrate component acts as a time-release reservoir, converting slowly to nitrite over the extended cure period.

Never substitute one for the other. Using #1 on a long dry-cured product depletes nitrite before the cure is complete. Using #2 on a quick-cure product can deliver excessive nitrite. Do not improvise ratios — insufficient nitrite does not protect against botulism; excessive nitrite is toxic.

Standard Dry Cure Formula (Bacon Belly — Per 5 lbs)

- Non-iodized kosher salt: 2 tablespoons
- Brown or white sugar: 1 tablespoon
- Prague Powder #1: 1 level teaspoon

Mix cure thoroughly before application — even distribution matters.

Application Method

1. Weigh meat; calculate cure ingredients per pound
2. Apply one-third of cure on Day 1 — rub into all surfaces including any skin folds, bone channels, or cuts
3. Place on a rack (so drainage can occur) skin-side down in a non-reactive pan; refrigerate at 38–40°F — never above 40°F during any stage of curing
4. Apply second one-third of cure at Day 3–4
5. Apply final one-third at Day 6–7
6. Curing time: 7 days per inch of meat thickness at the thickest point, or 1.5–2 days per pound. Whichever is longer.
7. How to tell when fully cured: Press the meat at its thickest point — it should feel uniformly firm with no soft spots. When sliced, the interior should be uniformly pink throughout.
8. After curing: Rinse off surface cure; rest (equalize) in refrigerator 24–48 hours at 38–40°F. Then smoke, age, or cook as required.

Salt Pork (Traditional Saturation Method)

Salt pork uses much more salt than equilibrium curing — the goal is saturation, not precision.

Salt ratio: 1 lb coarse non-iodized salt per 2–3 lbs of pork belly or fatback. A 1:2 salt-to-meat ratio is a common starting point; traditional recipes go higher.

Method:

1. Layer pork pieces and salt in a non-reactive container (ceramic crock, glass, or food-grade plastic), alternating generous salt layers. Each piece should be fully coated.
2. Store at 34–40°F throughout. A root cellar maintaining this temperature is the traditional environment.
3. After 4–5 days, the salt will have drawn considerable moisture from the meat, forming a brine (pickle) in the container. If meat is not fully submerged in this brine, prepare additional brine (3 cups coarse salt per gallon of water) to supplement.
4. Properly salted, fully submerged pork keeps 4–6 months at cellar temperatures.

Before use: Soak in cold water 12–24 hours, changing water every 4–6 hours, to remove excess salt. Taste-test a small cooked piece before seasoning dishes — salt pork is a seasoning element, not a standalone protein.

Method 4: Wet Brining and Immersion Curing

Wet brining submerges meat in a prepared salt solution, allowing cure to penetrate by osmosis and diffusion. Better than dry curing for penetrating bone-in cuts and very thick roasts.

Brine Concentrations

Strength	% Salt by weight	Approximate ratio
Light brine	3–5%	½ cup salt per gallon water
Medium brine	6–8%	¾–1 cup salt per gallon water
Curing brine (general)	10–12%	1¼ cups per gallon
Saturation (salt fish)	20–26%	2½–3 cups per gallon

For curing ham or corned beef: a 10–12% brine solution is standard. Process at 36–40°F throughout; 2–4 days per inch of thickness.

Injection Curing for Bone-In Cuts

Bone-in hams and thick roasts must be injection-cured — surface brining alone cannot penetrate to the center fast enough to prevent spoilage near the bone.

- Use a meat injector or curing pump
- Inject at multiple points 1–1.5 inches apart throughout the cut
- Target injected volume: approximately 10% of meat weight as brine
- After injection, continue with standard immersion brine for the cure period

Equilibrium Curing (Precision Method)

Traditional curing applies more salt than the meat absorbs, then removes or desalts. Equilibrium curing applies exactly the right amount — the meat absorbs all of it and needs no desalting.

How to calculate:

1. Decide target salt percentage in finished meat (typically 2–2.5% by weight)
2. Decide target nitrite (Prague Powder #1 or #2 — apply at 1 tsp per 5 lbs of meat weight regardless)
3. Weigh meat; calculate exact amounts of each ingredient
4. Apply and vacuum-seal or wrap tightly; refrigerate
5. Allow minimum 7 days, or 1–2 days per centimeter of thickness
6. Result: meat absorbs exactly what was applied; no desalting required

This is the most controlled method for the home curer. You cannot over-salt by leaving it too long.

Method 5: Smoking — Hot Smoking

Hot smoking at 225–275°F cooks and flavors meat simultaneously. It is not a standalone preservation method — smoked products require the same refrigeration as cooked food. The smoke contributes flavor and some antimicrobial surface compounds, but does not make food shelf-stable.

Temperature Targets

Chamber temperature: 225–275°F throughout.

Required internal temperatures (use a leave-in probe thermometer; do not rely on time alone):

Meat	Minimum internal temp
Poultry (whole or pieces)	165°F
Ground beef or pork	160°F
Beef steaks and roasts	145°F + 3-min rest
Whole pork	145°F + 3-min rest
Fish	145°F at thickest point
Brisket / pork shoulder (collagen breakd	195–205°F (above safety minimum)

Smoking Times by Meat

Meat	Smoker temp	Approximate time	Target internal
Beef brisket (whole)	225–250°F	1–1.5 hrs/lb (12–18 hrs total)	195–205°F
Pork shoulder / butt	225–250°F	1.5–2 hrs/lb (10–16 hrs)	195–205°F
Spare ribs	225–250°F	5–6 hours	195–205°F
Baby back ribs	225–250°F	4–5 hours	195–205°F
Whole chicken	275°F	2.5–4 hours	165°F
Whole turkey	275°F	30–45 min/lb	165°F
Fresh ham (uncured)	225–250°F	5–7 hours	145°F
Salmon fillet	200–225°F	2–4 hours	145°F
Fresh sausage links	225–250°F	2–3 hours	160°F (pork/beef)

Wood Selection

Wood	Flavor	Best for
Hickory	Strong, bacon-like, earthy	Pork, ribs, brisket, poultry
Mesquite	Intense, slightly bitter	Beef, wild game
Apple	Mild, sweet, fruity	Pork, poultry, fish
Cherry	Mild-medium, sweet	Pork, poultry, game birds
Alder	Very mild, clean	Fish (traditional), poultry
Pecan	Mild-medium, rich	Poultry, pork
Oak	Medium, clean	Beef, pork, lamb, fish
Maple	Mild, slightly sweet	Poultry, pork

Never use: pine, cedar, fir, spruce, elm, eucalyptus, treated lumber, or any pressed wood product. Resinous and chemically treated woods release toxic phenolic compounds into the smoke.

The Danger Zone During Long Smokes

Meat between 40°F and 140°F is in the bacterial growth zone. During a 12-hour brisket smoke, the meat spends several hours in this range while temperature slowly climbs. Total time in the danger zone must not exceed **4 hours.**

For any product that will spend more than 4 hours in the danger zone during smoking (particularly thick cuts at low temperatures), use Prague Powder #1 in a prior cure. The nitrite protects the anaerobic interior during the extended low-temperature phase.

Storage After Hot Smoking

Hot-smoked products are cooked food. Refrigerate within 2 hours. Consume within **3–4 days.** Freeze for longer storage (2–3 months for cured smoked products; 1–2 months for uncured).

Method 6: Cold Smoking Fish

Cold smoking operates at 68–86°F — temperatures that flavor but do not cook. Clostridium botulinum Type E is active at temperatures as low as 38°F. Without an adequate prior salt cure, cold-smoked fish is a botulism risk. This is the highest-risk method in this guide.

Cold-smoked fish stored at room temperature is not shelf-stable. It requires refrigeration at all times.

The Salt Cure Before Cold Smoking

Before cold smoking, fish must reach a **minimum 3.5% salt concentration in the water phase** of the flesh. This inhibits Type E botulinum growth. A target of 5% provides a better safety margin.

Wet brine method:

- Dissolve 1 cup non-iodized salt per quart of water (approximately 18–20% brine)
- Add ½–1 cup brown sugar per quart if desired
- Submerge fish completely; refrigerate
- Brine 8–12 hours for fillets under 1 inch thickness; 12–24 hours for thicker cuts
- Rinse thoroughly and pat dry

Dry cure method:

- Mix 1 part non-iodized salt to 3 parts brown sugar by volume (or equal parts for a saltier cure)
- Apply generously to all surfaces; refrigerate
- 8–24 hours depending on thickness

Developing the Pellicle

After brining and rinsing, the fish must develop a pellicle before smoking — a tacky, slightly glazed protein layer on the surface that smoke adheres to.

- Air-dry at room temperature for 1–2 hours; or
- In the refrigerator with a fan on low for 6–12 hours

Without a pellicle, smoke deposits unevenly and the surface stays moist rather than forming a protective layer.

Cold Smoking Protocol

- Chamber temperature: Below 90°F; traditionally 70–80°F; cooler if ambient temperature allows
- Duration: 8–24 hours for lox-style; 2–3 days for more heavily smoked product
- If ambient temperature rises above 40°F during a long cold smoking run, pause and return the fish to the refrigerator until conditions allow resuming

Canning After Cold Smoking (For Shelf Stability)

To make a shelf-stable cold-smoked fish product, pressure can it after smoking:

1. Lightly cold-smoke for flavor (not a full preservation smoke)
2. Pack vertically in pint jars, 1-inch headspace, no added liquid
3. Add exactly 4 quarts of cool tap water to the pressure canner
4. Process 110 minutes at 11 PSI dial-gauge (0–2,000 ft), 10 PSI weighted (0–1,000 ft)
5. Pint jars only; 16–22 quart canners only

Method 7: Jerky

Jerky is meat dried to low enough water activity that spoilage organisms cannot grow. The safety challenge is getting there without first creating conditions for pathogen growth — specifically, holding meat at temperatures that don't kill pathogens while still drying it adequately.

Meat Selection and Preparation

Best meats: Beef round or flank, venison, turkey breast. Trim all fat before drying — fat cannot be adequately dried and becomes rancid, dramatically shortening shelf life and producing off-flavors.

Slice thickness: No thicker than ¼ inch. Partially freeze meat (1–2 hours) to make consistent thin slicing easier.

Grain direction: With the grain produces chewy jerky; across the grain produces tender, slightly brittle jerky.

Pork and wild game — freeze first: Before processing, freeze pork or wild game at **0°F for a minimum of 30 continuous days.** This is the USDA-validated treatment for *Trichinella spiralis*. Alternatively, the post-dry oven heating step (see below) at 160°F also destroys trichinae — but this assumes the full interior reaches 160°F, which requires verification with a probe thermometer.

The 160°F Problem — And Two Solutions

Food dehydrators running at 130–155°F dry meat efficiently but may not bring the interior to 160°F. Evaporative cooling keeps the meat's surface temperature below the air temperature. Studies have confirmed that E. coli O157:H7 can survive dehydrator drying at 140°F. The danger zone is not addressed by drying time alone.

Solution A — Pre-heat method (preferred):
Place raw marinated meat strips and all marinade liquid in a pot. Bring to a full boil for 5 minutes. Drain immediately and place in dehydrator. This fully pasteurizes the meat before drying begins. Drying time is shorter. Texture is slightly paler before fully dry.

Solution B — Post-dry oven finish:
After jerky is fully dried to doneness (cracks but does not break when bent), spread strips on a baking sheet and heat in an oven at **275°F for 10 minutes.** Verify with an instant-read probe that the thickest pieces reach 160°F.

Both methods are validated. Choose based on your preference for texture and workflow.

Turkey and poultry jerky: Target 165°F internal, not 160°F. Apply the same pre-heat or post-heat protocol.

Drying Process

1. Apply marinade (see below); marinate 6–12 hours in refrigerator
2. Drain marinade; pat strips dry with paper towels
3. Arrange on dehydrator trays without touching or overlapping
4. Dry at 160°F (if your dehydrator reaches this temperature) for 4–8 hours; or at 140°F with pre-heat or post-heat method
5. Check doneness at the minimum time; continue as needed

Doneness test: Bend a cooled strip — it should **crack but not break in two.** A strip that bends without cracking is under-dried and will mold during storage. A strip that snaps in two is over-dried (quality issue, not safety).

Basic marinade (per 1–1.5 lbs of meat):

- ¼ cup soy sauce
- 1 tsp hickory-smoked salt or regular salt
- ½ tsp garlic powder
- ½ tsp black pepper
- ¼ tsp red pepper flakes (optional)

Salt in the marinade contributes flavor and minor water activity reduction, but does not replace the 160°F heat requirement.

Storage

Storage method	Duration
Room temperature, airtight container	2 weeks
Refrigerated	1–2 months
Frozen	6 months

Store in opaque containers — light accelerates fat oxidation.

Method 8: Rendering Fat — Lard, Tallow, and Schmaltz

Properly rendered animal fat is one of the most shelf-stable animal products achievable at home. With no residual moisture and minimal protein, it provides almost nothing for spoilage organisms to grow in.

Rendered fat also forms the basis of confit preservation (Method 9) and lard-sealed meat storage (traditional).

Fat Sources

Fat	Animal	Best cut for rendering
Lard	Pork	Leaf fat (from kidneys/loin) for finest
Tallow	Beef or mutton	Kidney fat (suet) for cleanest flavor
Schmaltz	Chicken or goose	Abdominal fat and skin

Leaf lard is the highest-quality pork fat — white, mild, nearly neutral flavor. Back fat produces stronger-flavored lard. Both are functionally equivalent for preservation purposes.

Dry Rendering (Preferred for Long-Term Storage)

Dry rendering produces a product with no residual water — the longest shelf life.

1. Chop or grind fat into pieces no larger than ½–1 inch
2. Place in a heavy pot or Dutch oven — no water added
3. Cook at 225–250°F in an oven, or on the stovetop on the lowest setting that maintains this temperature
4. Stir occasionally; cook 2–4 hours until fat is liquid and clear
5. Doneness: The cracklins (residual connective tissue) sink to the bottom and turn light golden-brown; the fat is still and clear; there is no bubbling or crackling sound (bubbling indicates residual moisture — continue until silent)
6. Strain immediately through cheesecloth into clean, sterile containers
7. Clarifying step for maximum shelf life: Return strained fat to a clean pot; heat briefly to 250°F while stirring gently. Complete silence at 250°F confirms all water has been driven off.

Wet Rendering

Same process with 1 cup of water per 5 lbs of fat added at the start. The water helps initially prevent scorching and is boiled off during rendering.

- Advantage: Less scorching risk; easier temperature control
- Disadvantage: Harder to confirm all water has been removed; slightly shorter shelf life

Storage

Condition	Duration
Room temperature, sealed, dark	3–6 months
Refrigerator, sealed	6–12 months
Freezer, sealed	1–2 years

Signs of rancidity: Soapy, sharp, or "crayon" odor. Any visible cloudiness or moisture in the jar (water contamination — re-render immediately or discard). Properly rendered fat has a clean, neutral to mildly fatty aroma.

Method 9: Confit

Confit cooks meat slowly in rendered fat at low temperature, then stores it submerged under the fat. The fat seals the surface from aerobic spoilage organisms. This is always a cooked, refrigerated product — not a room-temperature shelf-stable one.

What it works for: Duck legs, pork belly, pork shoulder, chicken thighs, rabbit, garlic.

Process

1. **Salt the meat 24–48 hours before:** Apply 1–1.5% of meat weight in salt (approximately ½–¾ tsp per pound) plus herbs and aromatics of choice. Refrigerate during this initial cure.
2. Rinse and pat dry before cooking. Wet meat causes oil to splatter and steam rather than confit.
3. Submerge in fat: Place meat in a heavy oven-safe pot; cover completely with duck fat, lard, or a combination.
4. Cook: Oven at 200–225°F. The fat temperature should be 180–200°F — a gentle, barely-moving simmer. Never allow the fat to boil.
5. Cooking time: Duck legs and chicken thighs: 2–3 hours. Pork shoulder or belly: 4–6 hours. Done when meat is very tender and pulls easily from bone; probe should read 160°F minimum for pork, 165°F for poultry.
6. Pack and seal: While everything is still hot, pack meat tightly into clean containers (Mason jars, ceramic crocks). Pour hot fat over to cover completely — no air pockets. Fat should cover the meat by at least ½ inch.
7. Cool to room temperature; refrigerate immediately.

Storage and Safety

Confit is not shelf-stable at room temperature. Traditional French larder storage worked because root cellars maintain 35–40°F year-round — effectively refrigerator temperature. A modern pantry or kitchen counter at 65–75°F is not safe for confit.

- Refrigerated, completely submerged in intact fat: 4–6 weeks
- Frozen: 6 months

Before using refrigerated confit: inspect the fat layer. If it has cracked or pulled away from the jar sides, exposing meat to air, treat it as regular refrigerated cooked meat and use within a week. The fat seal must be continuous and intact.

To serve: gently melt the fat in a warm oven until meat is accessible. The fat can be reused for the next confit batch.

Method 10: Bone Broth and Stock — Pressure Canning

Broth and stock are the most forgiving pressure-canned products — liquids heat through much faster than solid meat, and processing times are correspondingly shorter.

Preparation

Beef bone broth: Saw or crack large bones to expose marrow. Cover with water; simmer 3–4 hours covered. Cool completely. Skim and discard all solidified fat from the surface. Reheat to a boil before filling jars.

Poultry stock: Place carcass and bones in stockpot; cover with water; simmer 30–45 minutes until meat separates. Cool completely. Remove all solidified fat. Strain out solids. Reheat to boiling before filling jars.

Fat removal is mandatory. Fat on jar rims prevents sealing. Fat in a sealed jar can go rancid and compromise the product. Fat in the seal area can cause seal failure during storage. Remove it completely before canning.

Do not add starch, pasta, rice, flour, cream, or dairy before canning — these change density and heat penetration, and no tested recipe includes them.

Processing Times

Jar	Dial-gauge 0–2,000 ft	Weighted 0–1,000 ft
Pints	20 min / 11 PSI	20 min / 10 PSI
Quarts	25 min / 11 PSI	25 min / 10 PSI

Headspace: 1 inch. **Optional salt:** 1 tsp per quart, ½ tsp per pint.

Method 11: Salt Fish and Salt Cod

Traditional salt fish achieves shelf stability through two simultaneous mechanisms that must both be achieved: salt osmosis (raising internal salt concentration high enough to inhibit microbial growth) and drying (reducing water activity). One without the other produces a refrigerated product, not a shelf-stable one.

Target parameters for fully shelf-stable salt fish:

- Water phase salt concentration: ≥ 20%
- Water activity: ≤ 0.75
- Moisture content: ≤ 40%

Suitable Fish

Use lean white fish only: Cod, pollock, haddock, hake, cusk. Fatty fish (salmon, trout, mackerel) accumulate rancidity in the fat during the long curing and drying process. Lean fish have almost no fat — they cure and dry predictably.

Process

Salt phase (10–14 days):

1. Head and gut fish; split along the backbone to butterfly, or cut into fillets. Rinse in cold water; pat dry.
2. Layer fish and salt in a non-reactive container (wood, ceramic, or food-grade plastic — no metal). Salt ratio: at minimum 1 lb coarse non-iodized salt per 2 lbs of fish; traditional recipes use 1:1.
3. Skin side down; generous salt between layers; heavy salt layer on top. Refrigerate or store at 34–40°F.
4. After 24–48 hours, drain the expressed liquid (pickle). Remove fish; discard brine. Re-salt and restack. Repeat this every 2–3 days for the first week.
5. After 10–14 days, fish should be uniformly firm throughout — no soft spots or raw-feeling areas.

Drying phase (1–3 weeks, weather permitting):

1. Remove fish from salt; rinse surface; press under a weighted board to flatten.
2. Air-dry on elevated racks in a cool (40–60°F), dry location with good airflow. Traditional outdoor drying in cold, dry weather (Nordic and North Atlantic conditions) is ideal.
3. Turn fish daily.
4. Doneness: The fish is board-stiff throughout, completely rigid, leathery with a surface sheen. Knocking two pieces together produces a hollow wooden sound. No flexibility anywhere in the thickest section.

Storage: Properly dried salt fish keeps in a dry, dark, cool space (below 70°F) for **6–18 months.** Refrigeration extends this further.

Rehydration Before Use

Soak in cold water for **24–48 hours**, changing water every 8–12 hours with a minimum of 4 changes. Taste before cooking — if still very salty, soak longer. The fish is partially "cooked" by the salt cure and will be firm; it finishes in whatever application you cook it in.

Method 12: Pemmican

Pemmican is the most shelf-stable meat product achievable without any modern processing equipment. Used by Plains Indigenous peoples for centuries, by the fur trade, and by Arctic explorers as emergency rations, it requires no salt, no refrigeration, and no canning. Properly made, it is shelf-stable for years.

How It Works

Three mechanisms operating simultaneously:

1. Desiccation: Fully dried lean meat has water activity around 0.70 or below — below the threshold for pathogen or mold growth.
2. Fat encapsulation: Rendered tallow, poured hot over and mixed with dry meat powder, forms a continuous moisture barrier around every particle of protein.
3. No free water: The combination of dried meat and rendered fat leaves essentially no available water for microbial activity. No acidification, no salt, no heat treatment required for long-term stability.

Ingredients and Ratios

Lean meat: Bison, elk, deer, moose, or lean beef round — any very lean red meat. Remove every trace of fat before drying. Intramuscular fat will cause rancidity during storage.

Rendered tallow: Beef kidney suet is preferred for its neutral, firm character. Render fully (see Method 8).

Ratio: Equal parts by weight — **50% dried meat powder and 50% rendered tallow.**

Process

Step 1 — Dry the meat completely:
Slice lean meat as thin as possible (⅛ to ¼ inch). Dry at 145–160°F in a dehydrator until completely brittle — not chewy, not bendable. No moisture remaining anywhere.

Grind or pound dried meat into fine powder or small shreds. Traditional method was pounding between stones; a food processor or blender works efficiently. The finer the powder, the better the fat-to-protein mixing.

Step 2 — Render and clarify tallow:
Follow the dry rendering process in Method 8. Clarify fully — fat must be completely silent at 250°F before using.

Step 3 — Mix and form:

1. Heat tallow until fully liquid (200–225°F — hot but not smoking)
2. Pour hot tallow over dry meat powder; mix thoroughly until uniform
3. Optional traditional additions: dried berries (saskatoon, chokecherry, cranberry) add carbohydrates and antioxidants but reduce shelf life
4. Pour into molds (loaf pans lined with parchment work well)
5. Allow to solidify completely at room temperature
6. Cut into bars when solid

Storage

Condition	Duration
Cool (below 60°F), dark, dry	1–5 years
Warm or light-exposed	Shorter; fat oxidizes
With dried berries added	1–2 years

Signs of spoilage: Rancid fat smell (soapy, sharp). Visible mold (indicates water contamination or insufficiently dried meat — discard). Structural breakdown or unexpected softness.

Caloric density: approximately **500–600 calories per 100 grams** — among the highest of any food other than pure fat. Complete protein, complete fat, zero free water. It requires no cooking.

Sausage Making — Fresh and Cooked

Fresh Sausage (Bratwurst, Italian, Breakfast Links)

Fresh sausage contains no curing salts and is not shelf-stable. It must be refrigerated and cooked to 160°F (pork/beef) or 165°F (poultry) before eating.

Fat and lean ratio: 70–80% lean, 20–30% fat. Below 20% fat: dry and crumbly. Above 35%: greasy, won't bind well.

Salt: 1.5–2% of total meat weight. At minimum 1.5% for palatability and preservation; maximum 2.5%.

Refrigerator: 3–5 days raw. **Freezer:** 2–3 months.

Cooked and Smoked Sausage (Kielbasa, Hot Dogs)

Add Prague Powder #1 at 1 level teaspoon per 5 lbs of meat mixture. Smoke or cook to 160°F internal (165°F for poultry). Refrigerate; use within 1–2 weeks; freeze up to 2 months.

Fermented and Dry-Cured Sausage — A Warning

Home production of fermented dry sausage (salami, dry summer sausage, lonza) is the highest-risk activity in home meat preservation. Improperly made fermented sausage has caused botulism outbreaks. The anaerobic interior of stuffed sausage is exactly the environment C. botulinum requires, and botulinum toxin produces no visible or odor warning.

Fermented sausage is safe only when ALL of the following barriers are in place:

1. Prague Powder #2 at correct dosage
2. Acidification by starter cultures to pH ≤ 5.3, verified by pH meter within 72 hours of fermentation
3. Reduction of water activity to ≤ 0.91, verified by water activity meter
4. Adequate salt concentration (≥ 3% water phase salt) throughout

If any barrier fails, the product may contain botulinum toxin with no outward sign. This is not a beginner project. If you pursue it: get calibrated measurement equipment, use validated recipes from established sources (Pennsylvania State University Extension, USDA FSIS), and understand that there is no visual safety indicator.

Summer sausage made safe for beginners: Semi-dry summer sausage is made safe by cooking to 160°F internal after the fermentation stage. This cooking step eliminates the botulism risk and makes the process accessible without a calibrated curing chamber or water activity meter. The product is refrigerated after cooking and consumed within 3 weeks, not truly dry-cured shelf-stable sausage.

Part Three: Reference Tables

Quick Reference — Processing Times and Pressures

Product	Jar	Time	Dial-gauge (0–2k ft)	Weighted (0–1k ft)
Chicken, boneless	Pints	75 min	11 PSI	10 PSI
Chicken, boneless	Quarts	90 min	11 PSI	10 PSI
Chicken, bone-in	Pints	65 min	11 PSI	10 PSI
Chicken, bone-in	Quarts	75 min	11 PSI	10 PSI
Beef / pork / venison	Pints	75 min	11 PSI	10 PSI
Beef / pork / venison	Quarts	90 min	11 PSI	10 PSI
Ground meat	Pints	75 min	11 PSI	10 PSI
Ground meat	Quarts	90 min	11 PSI	10 PSI
Salmon / fatty fish	Pints only	100 min	11 PSI	10 PSI
Tuna (pre-cooked)	Pints only	100 min	11 PSI	10 PSI
Smoked fish	Pints only	110 min	11 PSI	10 PSI
Clams	Pints	70 min	11 PSI	10 PSI
Oysters	Pints	75 min	11 PSI	10 PSI
Meat stock / broth	Pints	20 min	11 PSI	10 PSI
Meat stock / broth	Quarts	25 min	11 PSI	10 PSI

Cure Ratios Quick Reference

Application	Salt	Sugar	Cure
Bacon dry cure (per 5 lbs)	2 Tbsp kosher	1 Tbsp	1 tsp Prague #1
Standard cure (per lb of meat)	0.25 oz (7g)	0.125 oz (3.5g)	0.04 oz (1.1g) #1 or #2
Salt pork	1 lb salt per 2–3 lbs pork	None required	Optional: 1 oz nitrate per 25 lbs
Corned beef brine	1¼ cup salt per gallon	½ cup	1 oz Prague #1 per gallon
Cold smoke brine (fish)	1 cup salt per quart water	½–1 cup	None (curing salts added for higher-risk
Gravlax cure (per lb salmon)	2 Tbsp coarse salt	1 Tbsp	None
Salt cod (per lb fish)	½–1 lb coarse salt	None	None
Pemmican	None	None	None (desiccation only)

Critical Temperature Reference

Temperature	What It Means
250°F (121°C)	Pressure canning maximum — destroys C. b
212°F (100°C)	Boiling water — insufficient for low-aci
205°F	Brisket / pork shoulder collagen breakdo
165°F	Required internal: poultry, turkey jerky
160°F	Required internal: ground beef/pork, jer
145°F + 3-min rest	Required internal: whole pork, lamb, bee

140°F	Dehydrator temperature for most jerky (w
40°F	Refrigerator — slows but does not stop b
38–40°F	Required curing temperature throughout a
0°F	Freezer — stops growth; 30 days at 0°F t

Storage Duration Quick Reference

Product / Method	Duration
Frozen meat (whole cuts)	6–12 months
Frozen ground meat	3–4 months
Frozen fish	3–6 months
Frozen cooked sausage	2–3 months
Pressure canned meat	12–18 months
Pressure canned fish	12–18 months
Pressure canned stock	12–18 months
Hot-smoked meat or fish	3–4 days refrigerated
Hot-smoked + vacuum sealed	Up to 2 weeks refrigerated
Jerky (room temp, airtight)	2 weeks
Jerky (refrigerated)	1–2 months
Salt pork (refrigerated in salt)	4–6 months
Salt cod (dried, shelf)	6–18 months
Gravlax (refrigerated)	5–7 days
Confit (refrigerated under fat)	4–6 weeks
Rendered lard/tallow (refrigerator)	6–12 months
Rendered lard/tallow (freezer)	1–2 years
Pemmican (cool and dark)	1–5 years

Critical Safety Reference

Botulism in cured meat: *Clostridium botulinum* thrives in the anaerobic (oxygen-free), protein-rich, low-acid interior of a thick cut of meat. Plain salt alone does not prevent it. Sodium nitrite (Prague Powder #1 or #2, depending on the cure) is the specific inhibitor. Any cured meat product intended for extended storage, room-temperature use, or smoking at low temperatures requires the appropriate curing salt at the correct dosage.

Botulinum Type E in fish: Unlike the strains associated with meat, Type E is cold-tolerant and can grow at temperatures as low as 38°F under the right conditions. Fish must be kept on ice from catch to processing. Cold-smoked fish requires adequate pre-cure (≥3.5% water phase salt) and continuous refrigeration.

Trichinella in pork and wild game: Freeze at 0°F for 30 continuous days (for pieces 6 inches thick or less), or cook to 160°F internal throughout (including in jerky via the post-dry oven method). Wild bear is higher-risk for trichinae than farm-raised pork; apply the same protocol.

Fermented sausage: No single safety indicator. All four barriers (nitrite, pH, water activity, salt) must be achieved simultaneously. This is not a beginner technique.

Never taste-test suspicious canned or cured meat. Botulinum toxin is colorless, odorless, and potentially lethal in quantities too small to detect. If something looks wrong, smells wrong, or shows any sign of compromise — discard without tasting.

Appendix: Authoritative Sources for Tested Recipes

Use tested recipes from these sources only. Do not adapt untested recipes for pressure-canned, cured, or cold-smoked animal products — the safety parameters have been established through laboratory testing that cannot be replicated by substitution or improvisation.

- National Center for Home Food Preservation — nchfp.uga.edu — all USDA-approved canning, curing, and smoking procedures
- USDA Complete Guide to Home Canning — free PDF from USDA; the foundational reference document
- Oregon State University Extension — PNW publications; particularly strong on fish smoking (PNW-238) and jerky (PNW-632)
- University of Missouri Extension — G2528 (bacon curing and processing)
- Pennsylvania State University Extension — sausage making and fermented meat guidance
- USDA FSIS (Food Safety and Inspection Service) — meat and poultry temperature and handling guidance
- Your state's cooperative extension service — locally relevant guidance; master food preserver programs

Related Seedwarden guides:

- Harvest Preservation Field Manual — plant-based preservation: vegetables, fruit, herbs, root cellaring
- Survival Garden Regional Plans — what to grow that you'll then need to preserve
- Seed Saving Field Manual — closing the loop on the full growing cycle

More from Seedwarden

If you found this guide useful, you might also like:

- Hunting, Fishing & Trapping Field Manual ($20) — Field skills for sourcing the wild game and fish you'll preserve
- Harvest Preservation Field Manual ($16) — Plant-side preservation: canning, drying, root cellaring for vegetables and fruit
- Small-Scale Livestock Field Manual ($18) — Raise chickens, rabbits, goats, and more on a quarter-acre or less

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